

Sex Differences in the Association Between Insulin Resistance and Systemic Inflammation Below the HbA1c Prediabetes Threshold: NHANES 2015–March 2020

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Abstract

Background: Insulin resistance and systemic inflammation are closely related features of metabolic dysfunction, but whether their association differs by sex before conventional glycemic thresholds for prediabetes are reached is not well established. We examined whether sex modifies the association between homeostatic model assessment of insulin resistance (HOMA-IR) and high-sensitivity C-reactive protein (hsCRP) among U.S. adults without diagnosed diabetes and with HbA1c < 5.7%.

Methods: We conducted a cross-sectional analysis of the 2015–March 2020 pre-pandemic National Health and Nutrition Examination Survey (NHANES). The primary analytic sample included 3,079 adults aged ≥ 20 years meeting the glycemic and fasting-subsample eligibility criteria. Survey-weighted linear regression modeled log-transformed hsCRP as a function of HOMA-IR, sex, their interaction, and period-specific adjustment for HbA1c, waist circumference, age, and non-HDL cholesterol. The primary inferential test was the survey-adjusted HOMA-IR-by-sex interaction. Sensitivity analyses evaluated exposure scale and upper-tail influence,

alternative adiposity specifications, selected clinical and behavioral factors, pregnancy, and restriction to $\text{hsCRP} \leq 10$ mg/L.

Results: The adjusted HOMA-IR–hsCRP association differed by sex in the primary analysis ($F_{1,40} = 9.05$, $p = 0.0045$). The estimated HOMA-IR slope was 0.0629 (95% CI, 0.0265 to 0.0992) among women and 0.0082 (95% CI, -0.0156 to 0.0319) among men, with a male-minus-female slope difference of -0.0547 (95% CI, -0.0914 to -0.0180). The direction of the sex contrast was maintained across alternative HOMA-IR scales and upper-tail restrictions, alternative adiposity specifications, and targeted smoking, statin, renal, hepatic, alcohol, and pregnancy sensitivities. However, restriction to $\text{hsCRP} \leq 10$ mg/L attenuated the male-minus-female slope difference by approximately 51%, to -0.0269 (95% CI, -0.0572 to 0.0033; $F_{1,40} = 3.24$, $p = 0.080$).

Conclusions: Among U.S. adults without diagnosed diabetes and with HbA1c below the conventional threshold for prediabetes, the adjusted association between HOMA-IR and hsCRP was more positive among females than among males. This sex contrast was stable across multiple metabolic and clinical sensitivity analyses but was substantially attenuated after restriction of the upper hsCRP range. These findings suggest that sex-related heterogeneity in metabolic–inflammatory coupling may be detectable below conventional glycemic thresholds, while also indicating that the magnitude of this heterogeneity depends importantly on the inflammatory range represented in the analytic population.

1 Introduction

Insulin resistance and systemic inflammation are closely related features of metabolic dysfunction. Higher concentrations of C-reactive protein (CRP) have been associated with greater insulin resistance in population-based studies of adults without diabetes, including nationally representative U.S. data [Meng et al., 2007]. Longitudinal evidence from the Coronary Artery Risk Development in Young Adults (CARDIA) study has further demonstrated associations between CRP and both concurrent and subsequent HOMA-IR, although much of the association was attenuated after accounting for adiposity [Park et al., 2009]. Similar associations between CRP and HOMA-IR have been reported in other nondiabetic populations and have persisted after adjustment for abdominal adiposity [Chou et al., 2010]. These observations suggest that inflammatory and insulin-resistance phenotypes may be coupled before the development of overt diabetes.

Sex may be an important source of heterogeneity in this relationship. Women and men differ in insulin sensitivity and in the distribution and metabolic function of adipose tissue, with sex hormones and inflammatory pathways among the mechanisms proposed to contribute to these differences [Gado et al., 2024]. Immune and inflammatory responses also differ by sex across both innate and adaptive pathways [Klein and Flanagan, 2016]. Consistent with these physiological differences, women have generally been reported to have higher CRP concentrations than men in population-based studies [Lakoski et al., 2006], and the association between total fat mass and CRP may itself differ by sex [Khera et al., 2009]. Thus, sex differences in average concentrations of metabolic or inflammatory markers may not fully characterize sex-related heterogeneity in the

relationship between these systems.

Previous studies have examined the relationship between HOMA-IR and CRP in women and men, but evidence regarding sex-specific coupling is limited and not uniform. Sex-stratified analyses in nondiabetic Taiwanese adults found positive adjusted HOMA-IR–CRP associations in both women and men [Chou et al., 2010], whereas CARDIA reported no significant interaction between CRP and sex in the prediction of HOMA-IR [Park et al., 2009]. More recently, analysis of a representative Canadian adult population explicitly examined potential differences by sex and reported differing patterns of association between CRP and HOMA-IR in women and men [Shahid et al., 2023]. Less is known about whether the adjusted association between insulin resistance and systemic inflammation differs by sex among adults who remain below conventional glycemic thresholds for prediabetes. This distinction may be informative because clinical glycemic categories do not necessarily imply physiological homogeneity in insulin resistance or inflammatory state.

We therefore examined whether sex modifies the association between HOMA-IR and high-sensitivity C-reactive protein (hsCRP) among U.S. adults aged 20 years or older without diagnosed diabetes and with HbA1c < 5.7%, using data from the 2015–March 2020 pre-pandemic National Health and Nutrition Examination Survey (NHANES). The primary analysis formally tested the HOMA-IR-by-sex interaction while accounting for the complex survey design and relevant metabolic covariates. We further evaluated the stability of the estimated sex difference across alternative HOMA-IR scales and exposure distributions, multiple representations of adiposity, selected clinical and behavioral factors, pregnancy, and restriction of the upper hsCRP range.

2 Methods

2.1 Study Design and Data Source

We conducted a cross-sectional analysis of publicly available data from the National Health and Nutrition Examination Survey (NHANES), using the 2015–2016 cycle and the 2017–March 2020 pre-pandemic data release. NHANES is a nationally representative survey of the noninstitutionalized U.S. population that combines interviews, physical examinations, and laboratory measurements using a complex, multistage probability sampling design [Akinbami et al., 2022].

The 2015–2016 and 2017–March 2020 datasets were harmonized at the participant level using the NHANES participant identifier (SEQN). Variables required for the primary analysis were obtained from the demographic, examination, diabetes questionnaire, glycohemoglobin, fasting glucose, fasting insulin, high-sensitivity C-reactive protein, and lipid files. Additional questionnaire and laboratory files were used for sensitivity analyses. Because HOMA-IR requires both fasting glucose and fasting insulin, analyses were conducted within the NHANES fasting laboratory subsample and used the corresponding fasting-subsample survey weights.

2.2 Analytic Population

The analytic population was restricted to adults aged 20 years or older who had no self-reported diagnosis of diabetes and had HbA1c $< 5.7\%$, the lower boundary of the conventional HbA1c-defined prediabetes range [[American Diabetes Association Professional Practice Committee for Diabetes, 2026](#)]. Participants were additionally required to belong to the qualifying NHANES fasting laboratory subsample, as indicated by a nonzero fasting-subsample survey weight, and to have valid fasting glucose and fasting insulin measurements, positive HOMA-IR and hsCRP values, and complete data for variables included in the primary model. No imputation of missing data was performed.

Fasting-subsample eligibility was independently verified using participant-reported fasting duration from the NHANES fasting questionnaire, calculated as $\text{PHAFSTHR} + \text{PHAFSTMN}/60$ [[National Center for Health Statistics, 2017, 2021](#)]. All participants with a qualifying fasting-subsample weight in the final analytic population reported fasting durations within the NHANES qualifying range of 8 to < 24 hours.

Of 3,182 participants meeting the remaining eligibility and measurement requirements, 103 (3.2%) were excluded because of incomplete primary-model covariate data, yielding a final primary analytic sample of 3,079 participants. The final sample represented approximately 144.3 million U.S. adults after application of the combined survey weights.

No a priori sample-size calculation was performed; the analytic sample comprised all participants from the specified NHANES survey periods who met the study eligibility criteria and had the data required for the primary analysis.

The HbA1c restriction was used to examine metabolic heterogeneity specifically among individuals remaining below the conventional laboratory threshold for prediabetes. This restriction defines the population under study and should not be interpreted as establishing that all included participants were otherwise metabolically healthy.

2.3 Exposure: HOMA-IR

The primary exposure was the Homeostatic Model Assessment of Insulin Resistance (HOMA-IR), a surrogate measure of insulin resistance derived from fasting insulin and fasting glucose concentrations [[Matthews et al., 1985, Wallace et al., 2004](#)]. HOMA-IR was calculated as

$$\text{HOMA-IR} = \frac{\text{fasting insulin } (\mu\text{U/mL}) \times \text{fasting glucose } (\text{mg/dL})}{405}.$$

HOMA-IR was modeled continuously and retained on its original scale in the primary analysis. Raw HOMA-IR was retained to preserve direct interpretation per one-unit difference and continuity with

the preceding analysis of this NHANES cohort. Because HOMA-IR was right-skewed, natural-log-transformed HOMA-IR and progressively trimmed upper-tail samples were evaluated in sensitivity analyses.

2.4 Outcome: hsCRP

The primary outcome was high-sensitivity C-reactive protein (hsCRP), modeled as its natural logarithm,

$$Y = \ln(\text{hsCRP}).$$

The log transformation was retained from the preceding analysis of this cohort because raw hsCRP was strongly right-skewed and log transformation substantially improved regression residual behavior. For models using $\ln(\text{hsCRP})$, coefficients may be expressed as proportional differences on the original hsCRP scale according to

$$100(e^\beta - 1)\%.$$

A sensitivity analysis specified before examination of the corresponding Study #2 result restricted the analytic population to participants with $\text{hsCRP} \leq 10$ mg/L to evaluate dependence of the estimated sex interaction on the upper range of the hsCRP distribution.

2.5 Sex and Primary Covariates

Sex was the focal effect modifier and was coded from the NHANES sex variable as female (reference category) or male. The primary analysis therefore tested whether the adjusted HOMA-IR–hsCRP slope differed between females and males; sex was not included as a separate ordinary adjustment covariate outside the interaction term.

The primary harmonized adjustment set consisted of HbA1c, waist circumference, continuous age, and non-HDL cholesterol. Non-HDL cholesterol was calculated as

$$\text{non-HDL} = \text{total cholesterol} - \text{HDL cholesterol}.$$

Waist circumference was retained as the primary adiposity measure. Alternative adiposity specifications using body mass index (BMI) and waist-to-height ratio, as well as a model without adiposity adjustment, were evaluated as sensitivity analyses.

2.6 NHANES Survey Design and Weighting

All population estimates and inferential analyses incorporated the NHANES complex sampling design [Akinbami et al., 2022]. Because calculation of HOMA-IR required both fasting glucose and fasting insulin, analyses used the corresponding fasting-subsample survey weights rather than general examination weights.

Survey variance estimation incorporated the NHANES masked variance stratum (SDMVSTRA) and masked variance unit (SDMVPSU) variables. For the combined 2015–March 2020 analysis, fasting weights were adjusted for the unequal durations represented by the component datasets according to NHANES guidance [Akinbami et al., 2022]. The 2015–2016 fasting weight was rescaled as

$$W_{15/16}^* = \text{WTS AF2YR} \left(\frac{2}{5.2} \right),$$

and the 2017–March 2020 pre-pandemic fasting weight as

$$W_{17/20}^* = \text{WTS AFPRP} \left(\frac{3.2}{5.2} \right).$$

Masked variance strata and variance units were uniquely identified across periods before the datasets were combined. The resulting combined design provided 40 survey design degrees of freedom. Survey-design-based variance estimates were used for standard errors, confidence intervals, t -tests, and F -tests; participants were not treated as independent simple-random-sample observations.

Analyses were performed on the full survey design while defining the eligible analytic population as a survey domain. Contributions to estimating equations from observations outside the analytic domain were set to zero, preserving the design structure for domain variance estimation.

2.7 Primary Statistical Analysis

The primary analysis estimated the adjusted association between HOMA-IR and log-transformed hsCRP and formally tested whether that association differed by sex. The Study #2 primary interaction model was

$$\ln(\text{hsCRP}) \sim \text{HOMA-IR} \times \text{sex} + \text{period} \times (\text{HbA1c} + \text{waist} + \text{age} + \text{non-HDL}).$$

Equivalently, the focal portion of the model may be represented as

$$\ln(\text{hsCRP}_i) = \beta_0 + \beta_1 H_i + \beta_2 M_i + \beta_3 (H_i M_i) + \text{period-specific adjustment terms} + \epsilon_i,$$

where H_i denotes HOMA-IR and M_i is an indicator for male sex. With female as the reference category, β_1 represents the adjusted female HOMA-IR slope and β_3 represents the male-minus-female difference in HOMA-IR slopes. The male slope is therefore $\beta_1 + \beta_3$.

The primary inferential test was the survey-adjusted Wald test of

$$H_0 : \beta_3 = 0.$$

Evidence for effect modification was therefore determined from the formal HOMA-IR-by-sex interaction rather than by comparing whether the female and male sex-specific coefficients separately crossed a conventional statistical significance threshold. Sex-specific slopes, the male-minus-female slope contrast, and corresponding 95% confidence intervals were estimated from the same fitted interaction model.

A survey-period indicator was included, and HbA1c, waist circumference, continuous age, and non-HDL cholesterol were permitted to have period-specific coefficients through interactions with survey period. The HOMA-IR-by-sex association remained the common effect-modification structure of primary interest. This specification avoided assuming that adjustment-variable relationships with hsCRP were identical in the 2015–2016 and 2017–March 2020 components.

2.8 Sensitivity Analyses

Sensitivity analyses were organized according to specific potential sources of inferential vulnerability and were treated as robustness analyses rather than alternative primary models.

Exposure scale and upper-tail influence. To assess whether the sex interaction depended on the right-skewed raw HOMA-IR scale or a small number of extreme exposure values, the primary model was repeated using natural-log-transformed HOMA-IR and after excluding observations above the 99th, 97.5th, and 95th percentiles of the HOMA-IR distribution. Residual diagnostics for the raw- and log-HOMA-IR specifications were also examined.

Adiposity specification. The primary waist-circumference model was compared with specifications using BMI, waist-to-height ratio, and no adiposity adjustment. Both available-case and common-complete-case analyses were used to distinguish changes associated with adiposity specification from those caused by differential missingness.

Additional adjustment sensitivities. Targeted sensitivity analyses evaluated additional adjustment for smoking status, statin use, renal function, hepatic/metabolic biochemistry, and alcohol-use

status. Smoking was categorized as never, former, or current. Statin use was identified from reported prescription medications during the preceding 30 days using NHANES drug classifications for HMG-CoA reductase inhibitors. Renal sensitivity analyses used continuous 2021 CKD-EPI creatinine-based estimated glomerular filtration rate (eGFR) [Inker et al., 2021], with log-transformed serum creatinine as a complementary specification. Log-transformed alanine aminotransferase (ALT) was used for the hepatic/metabolic sensitivity. Alcohol use was reconstructed across the two questionnaire structures as zero past-year exposure, very-low lifetime exposure, or positive past-year drinking frequency. For each sensitivity with missing exposure information, the reference and expanded models were compared within the same common sample.

Current pregnancy. Because current pregnancy represents a physiologically distinct metabolic and inflammatory state, the primary model was repeated after excluding participants with confirmed current pregnancy. A complementary analysis additionally excluded participants whose pregnancy status could not be determined. These analyses addressed current pregnancy only and were not interpreted as accounting for menopausal or other reproductive states.

Upper hsCRP restriction. The primary model was repeated after restriction to $\text{hsCRP} \leq 10$ mg/L. Elevations above 10 mg/L have historically received particular attention in the interpretation of CRP, although such elevations need not represent acute inflammation and may be repeatedly observed in some individuals [Ishii et al., 2012]. This threshold had been used in the preceding analysis of the same cohort and was not selected from the Study #2 interaction results. No alternative hsCRP thresholds were explored after observing this sensitivity analysis.

2.9 Descriptive Follow-up of the Upper hsCRP Range

Because restriction to $\text{hsCRP} \leq 10$ mg/L materially attenuated the estimated HOMA-IR-by-sex interaction, a descriptive follow-up was conducted to characterize participants with $\text{hsCRP} > 10$ mg/L. This analysis was undertaken to understand the sensitivity result and was not used to define new exclusion criteria or alternative primary models.

A limited complete-blood-count panel was specified before examining its values: total white blood cell count, absolute neutrophil count, neutrophil percentage, absolute lymphocyte count, absolute monocyte count, and a derived neutrophil-to-lymphocyte ratio. These measures were summarized descriptively by sex and hsCRP group. The HOMA-IR distribution was likewise summarized within the four sex-by-hsCRP groups. Scatterplots of raw HOMA-IR against log-hsCRP were examined separately by sex using both the full HOMA-IR range and a HOMA-IR 0–12 display window; the latter was a visualization restriction only and did not exclude observations from any analysis.

The descriptive CBC analysis was not used to classify participants as having acute infection.

Similarly, the joint HOMA-IR/hsCRP patterns were interpreted as descriptive features of the observed distribution rather than as evidence of a causal biological mechanism.

2.10 Deferred Reproductive-State Analysis

Reproductive-health questionnaire variables describing recent menses, hysterectomy, and bilateral oophorectomy were examined as predictor-only quality-control work after the sex interaction had been identified. The available cross-sectional information did not provide a uniquely determined menopausal classification for all women, and no HOMA-IR–hsCRP outcome model was fitted using post hoc reproductive-state categories. Reproductive and menopausal physiology was therefore deferred as a separate future research question rather than added as another Study #2 sensitivity analysis.

2.11 Software and Reproducibility

Analyses were performed in Python using NumPy, pandas, Patsy, SciPy, and statsmodels. Survey-weighted linear regression used Taylor-linearized design-based covariance estimation with NHANES strata, primary sampling units, fasting-subsample weights, and domain score-zeroing. The validated survey analysis implementation and cohort-construction framework from the preceding NHANES study were retained without modification for the present analysis.

All sensitivity analyses were documented sequentially in an analysis log, including the specification of exposure definitions and analytic decisions before outcome-model fitting where applicable. Analyses were frozen after completion; additional thresholds or model variants were not selected to recover statistical significance.

2.12 Use of Generative Artificial Intelligence

Generative artificial intelligence (ChatGPT, OpenAI) was used as a research-support tool during this project, including assistance with drafting and revising portions of the manuscript and with writing, optimizing, and debugging portions of the reproducible Python analysis pipeline. The authors determined the research questions, analytic strategy, statistical methods, interpretation of results, and conclusions. AI-assisted analytical code and outputs were subject to human review and validation against the specified analytic framework, mathematical expectations, and NHANES survey-design constraints. All manuscript content was critically reviewed and approved by the authors, who take full responsibility for the accuracy, integrity, interpretation, and conclusions of the work.

3 Results

3.1 Analytic Population

The final primary analytic sample included 3,079 adults aged 20 years or older without diagnosed diabetes and with HbA1c $< 5.7\%$. After application of the combined NHANES fasting-subsample weights, this sample represented approximately 144.3 million U.S. adults. The complex survey design provided 40 denominator degrees of freedom for design-based inference. Sex-stratified descriptive characteristics are shown in Table 1.

Table 1: Characteristics of the primary analytic population by sex

Characteristic	Female ($n = 1,638$)	Male ($n = 1,441$)
Weighted population, millions	74.2	70.2
Age, years	43.6 (0.6)	42.0 (0.6)
HbA1c, %	5.26 (0.01)	5.25 (0.01)
Waist circumference, cm	94.4 (0.6)	99.2 (0.6)
Non-HDL cholesterol, mg/dL	126.6 (1.4)	135.1 (1.6)
Fasting glucose, mg/dL	97.3 (0.3)	101.5 (0.4)
Fasting insulin, $\mu\text{U/mL}$	7.76 (5.25–12.01)	7.86 (5.01–13.74)
HOMA-IR	1.87 (1.21–2.88)	2.00 (1.25–3.43)
hsCRP, mg/L	1.70 (0.67–3.93)	1.30 (0.61–2.67)

Values are survey-weighted mean (SE) for age, HbA1c, waist circumference, non-HDL cholesterol, and fasting glucose, and survey-weighted median (IQR) for fasting insulin, HOMA-IR, and hsCRP.

The analytic population included adults aged ≥ 20 years without diagnosed diabetes and with HbA1c $< 5.7\%$, meeting the fasting-subsample and complete-case criteria for the primary model.

Weighted population estimates use the combined NHANES 2015–2016 and 2017–March 2020 pre-pandemic fasting-subsample weights.

No hypothesis tests comparing baseline characteristics by sex were performed.

3.2 Primary HOMA-IR-by-Sex Interaction

In the locked primary model, the adjusted association between HOMA-IR and log-transformed hsCRP differed by sex ($F(1,40) = 9.05$, $p = 0.0045$; Table 2). The estimated HOMA-IR slope among females was 0.0629 (SE 0.0180; 95% CI 0.0265 to 0.0992), whereas the estimated slope among males was 0.0082 (SE 0.0117; 95% CI -0.0156 to 0.0319). The male-minus-female difference in slopes was -0.0547 (95% CI -0.0914 to -0.0180).

On the original hsCRP scale, these coefficients correspond descriptively to approximately a 6.5% increase in hsCRP per one-unit higher HOMA-IR among females and a 0.8% increase among males, conditional on the fitted model. Inference, however, was based on the log-hsCRP scale and on the formal HOMA-IR-by-sex interaction rather than on separate tests of the two sex-specific slopes.

Table 2: Primary survey-weighted HOMA-IR-by-sex interaction model

Estimate	β	SE	95% CI	p
Female HOMA-IR slope	0.0629	0.0180	0.0265 to 0.0992	0.0012
Male HOMA-IR slope	0.0082	0.0117	−0.0156 to 0.0319	0.4905
Male-minus-female slope contrast	−0.0547	—	−0.0914 to −0.0180	0.0045

Outcome: $\ln(\text{hsCRP})$; exposure: raw continuous HOMA-IR. The model included $\text{HOMA-IR} \times \text{sex}$ and period-specific adjustment for continuous HbA1c, waist circumference, age, and non-HDL cholesterol. Female was the reference sex. The formal interaction test for equality of the sex-specific HOMA-IR slopes was $F(1, 40) = 9.053$, $p = 0.00452$. Inference incorporated the NHANES fasting-subsample weights, masked strata, and masked PSUs. Analytic $n = 3,079$.

3.3 Exposure Scale and Upper-Tail Influence

The HOMA-IR distribution was right-skewed (median 1.99; 95th percentile 6.66; 97.5th percentile 8.44; 99th percentile 11.31; maximum 38.97). The estimated sex contrast persisted in the same direction after natural-log transformation of HOMA-IR and after progressive restriction of the upper HOMA-IR tail (Table S1). In the full sample, the log-HOMA-IR interaction was $F(1, 40) = 10.88$ ($p = 0.0021$). Under raw-HOMA-IR restrictions at the 99th, 97.5th, and 95th percentiles, male-minus-female slope contrasts were -0.0735 , -0.1036 , and -0.1001 , respectively, compared with -0.0547 in the full primary sample. Residual diagnostics did not identify a qualitative deficiency requiring replacement of the primary raw-HOMA-IR specification.

3.4 Adiposity Sensitivity

Adiposity adjustment substantially attenuated the HOMA-IR–hsCRP slopes relative to a model without adiposity adjustment, but the estimated sex contrast remained in the same direction across BMI, waist circumference, and waist-to-height ratio specifications (Table S2). The male-minus-female contrast was -0.0955 (95% CI -0.1616 to -0.0293) without adiposity adjustment, -0.0423 (95% CI -0.0811 to -0.0035) with BMI, -0.0547 (95% CI -0.0914 to -0.0180) with the primary waist circumference specification, and -0.0416 (95% CI -0.0791 to -0.0041) with waist-to-height ratio. Common-complete-case analyses ($n = 3,072$) produced nearly identical estimates, indicating that differences among these specifications were not attributable to differential missingness.

3.5 Additional Adjustment and Exclusion Sensitivities

Targeted adjustment for smoking, current prescription statin use, renal function, log-transformed ALT, and alcohol-use status produced little change in the estimated HOMA-IR-by-sex contrast within their respective common samples (Table S3). The alcohol analysis was particularly informative about sample restriction: among the 2,919 participants with classifiable alcohol status, the reference contrast was -0.0673 (95% CI -0.1060 to -0.0287), compared with -0.0665 (95% CI -0.1047 to

-0.0284) after alcohol adjustment. Thus, the difference between the full-cohort primary estimate and the alcohol-sample estimate reflected sample restriction rather than alcohol adjustment.

Exclusion of 48 participants with confirmed current pregnancy likewise produced little change ($n = 3,031$; male-minus-female contrast -0.0567 ; $F(1,40) = 9.04$, $p = 0.0045$). Excluding both confirmed pregnancy and the 15 participants whose pregnancy status could not be determined produced a nearly identical contrast (-0.0567 ; $n = 3,016$).

3.6 Sensitivity to the Upper hsCRP Range

Restriction to participants with $\text{hsCRP} \leq 10$ mg/L materially attenuated the estimated sex difference. In the full primary sample, the male-minus-female HOMA-IR slope contrast was -0.0547 (95% CI -0.0914 to -0.0180). Among the 2,895 participants with $\text{hsCRP} \leq 10$ mg/L, the female slope decreased from 0.0629 to 0.0438, whereas the male slope increased from 0.0082 to 0.0168. The resulting male-minus-female contrast was -0.0269 (95% CI -0.0572 to 0.0033), with $F(1,40) = 3.24$ ($p = 0.0796$).

Thus, restriction to $\text{hsCRP} \leq 10$ mg/L reduced the absolute magnitude of the interaction contrast by approximately 50.8%. The attenuation therefore reflected convergence of both sex-specific slopes rather than only a change in whether the interaction crossed a conventional statistical-significance threshold.

3.7 Descriptive Characterization of $\text{hsCRP} > 10$ mg/L

Of the 184 participants with $\text{hsCRP} > 10$ mg/L, 128 were female and 56 were male, corresponding to unweighted proportions of 7.81% and 3.89% within the female and male analytic samples, respectively. Complete-blood-count measures were available for all 184 participants.

In both sexes, the $\text{hsCRP} > 10$ mg/L group showed descriptively higher WBC, absolute neutrophil count, neutrophil percentage, and neutrophil-to-lymphocyte ratio than the $\text{hsCRP} \leq 10$ mg/L group (Table S4). The HOMA-IR distribution showed a more pronounced shift among females: median HOMA-IR was 3.01 versus 1.91 for $\text{hsCRP} > 10$ versus ≤ 10 mg/L, with corresponding 90th percentiles of 7.40 and 4.51. Among males, median HOMA-IR was similar between hsCRP groups (2.04 versus 2.05), although the upper distribution extended farther among those with $\text{hsCRP} > 10$ mg/L (90th percentile 6.77 versus 5.13).

Sex-stratified scatterplots were consistent with these descriptive summaries: female observations with $\text{hsCRP} > 10$ mg/L extended farther into the higher-HOMA-IR portion of the joint distribution, whereas numerous male observations with higher HOMA-IR remained below the 10 mg/L hsCRP threshold. These descriptive analyses were not used to classify acute infection or to define additional exclusions.

4 Discussion

In this nationally representative sample of U.S. adults without diagnosed diabetes and with HbA1c below the conventional threshold for prediabetes, the adjusted association between HOMA-IR and hsCRP differed by sex. The association was more positive among women than men, with a male-minus-female difference in HOMA-IR slope of -0.0547 (95% CI, -0.0914 to -0.0180). This contrast persisted across alternative HOMA-IR scales, progressively trimmed exposure distributions, alternative representations of adiposity, and multiple targeted clinical and behavioral sensitivity analyses. The principal qualification was the restriction to $\text{hsCRP} \leq 10$ mg/L, which reduced the estimated sex contrast by approximately 51%. Thus, the findings support sex-related heterogeneity in the relationship between fasting insulin resistance and systemic inflammation in this metabolically selected population, while indicating that the magnitude of this heterogeneity is substantially influenced by the upper range of the hsCRP distribution.

The observed sex difference is biologically plausible in the context of established sexual dimorphism in metabolic and immune physiology. Sex differences have been described in insulin sensitivity and resistance, with potential contributions from body-fat distribution and function, inflammatory biology, and sex hormones [Gado et al., 2024]. Immune responses likewise differ by sex across both innate and adaptive pathways and vary across the life course and reproductive state [Klein and Flanagan, 2016]. At the population level, women have been reported to have higher CRP concentrations than men even after accounting for several relevant covariates [Lakoski et al., 2006], and the association between total fat mass and log-transformed CRP has been reported to be steeper among women than men [Khera et al., 2009]. These observations provide biological and epidemiologic context for the present finding, but they do not establish the mechanism underlying the sex difference in HOMA-IR–hsCRP coupling observed here.

Adiposity appears to be an important component of this relationship. Adipose tissue is metabolically and immunologically active, with adipose-derived signals contributing to both insulin sensitivity and systemic inflammatory biology [Ouchi et al., 2011]. In the present analysis, omission of adiposity adjustment substantially increased both sex-specific HOMA-IR slopes and the estimated sex contrast, whereas adjustment using BMI, waist circumference, or waist-to-height ratio attenuated the contrast but did not eliminate it. This pattern is consistent with adiposity accounting statistically for a meaningful portion of the observed HOMA-IR–hsCRP relationship, while the persistence of the sex contrast across several anthropometric specifications suggests that the finding is not attributable to any single representation of adiposity. Prior population-based work has similarly reported a steeper association between total fat mass and CRP among women than men [Khera et al., 2009]. Because anthropometric measures are imperfect proxies for body composition and fat distribution, however, these analyses should not be interpreted as demonstrating an association independent of adipose biology.

The estimated sex contrast was also largely unchanged across several targeted sensitivity analyses

addressing smoking, statin use, renal function, hepatic status, alcohol use, and pregnancy. These factors were evaluated because each could plausibly influence metabolic or inflammatory measurements or their interpretation. Their inclusion or exclusion produced little change in the estimated HOMA-IR–hsCRP sex interaction. These analyses therefore reduce concern that the primary finding is readily explained by any one of these measured factors, but they do not eliminate residual confounding or establish that these factors are biologically irrelevant to the observed association. For alcohol, comparison with a reference model fitted to the same classifiable sample further indicated that the apparent difference from the full-cohort estimate reflected sample restriction rather than alcohol adjustment itself.

The most important qualification to the primary finding emerged when the analysis was restricted to participants with $\text{hsCRP} \leq 10 \text{ mg/L}$. The estimated sex contrast was attenuated by approximately 51%, from -0.0547 in the full analytic population to -0.0269 after restriction, and the confidence interval for the restricted estimate included zero. The 10-mg/L threshold has commonly been used in epidemiologic studies to reduce the potential influence of acute inflammatory states when CRP is used as a marker of chronic low-grade inflammation. However, values above this threshold cannot be assumed to represent acute infection. Repeated-measure studies have demonstrated that CRP concentrations above 10 mg/L may persist, particularly among women and individuals with obesity, suggesting that exclusion of all such observations can also remove participants with sustained inflammatory phenotypes [Ishii et al., 2012]. Accordingly, the attenuation observed here should not be interpreted simply as elimination of acute inflammatory outliers. Rather, it indicates that the estimated sex difference in HOMA-IR–hsCRP coupling is substantially influenced by participants in the upper range of the hsCRP distribution.

Descriptive evaluation of participants with $\text{hsCRP} > 10 \text{ mg/L}$ provided additional context for this sensitivity. In both sexes, participants in the upper hsCRP range had higher typical leukocyte counts, absolute neutrophil counts, neutrophil percentages, and neutrophil-to-lymphocyte ratios than participants with $\text{hsCRP} \leq 10 \text{ mg/L}$, consistent with greater inflammatory or immune activation at the group level. The distribution of HOMA-IR, however, differed more distinctly among women: median HOMA-IR was substantially higher among women with $\text{hsCRP} > 10 \text{ mg/L}$ than among women below the threshold, whereas the corresponding male medians were nearly identical. This pattern helps explain the geometry of the sensitivity result: observations in the upper inflammatory range among women were also shifted toward higher HOMA-IR, while the analogous male observations showed less evidence of a comparable metabolic shift. These descriptive findings do not establish a biological mechanism, but they indicate that the attenuation after hsCRP restriction reflects structure in the joint HOMA-IR–hsCRP distribution rather than merely the removal of a small number of arbitrarily defined extreme observations.

Reproductive physiology represents an important potential direction for subsequent investigation. Sex differences in insulin sensitivity vary across the life course, and the relative metabolic advantage observed among women before menopause appears to diminish after the menopausal transition;

sex hormones, adipose distribution, and inflammatory signaling have all been proposed as contributing mechanisms [Gado et al., 2024, Henstridge et al., 2019]. However, reproductive state could not be classified reliably from age or recent menstruation alone in the present NHANES sample. Hysterectomy, bilateral oophorectomy, natural menopause, and exogenous hormone use represent physiologically distinct states that would require explicit phenotype definitions and separate consideration. We therefore did not perform post hoc outcome modeling by menopausal or reproductive category. Whether the observed sex difference in HOMA-IR–hsCRP coupling varies across reproductive states remains an important question for future study.

Several limitations should be considered when interpreting these findings. First, the cross-sectional design precludes determination of temporal or causal direction between insulin resistance and systemic inflammation. Second, HOMA-IR is a surrogate derived from fasting glucose and insulin rather than a direct measure of insulin sensitivity such as the hyperinsulinemic-euglycemic clamp [Matthews et al., 1985, Wallace et al., 2004]. This distinction may be particularly relevant to the present study because relationships between fasting glucose–insulin indices and clamp-measured insulin sensitivity can vary by sex and other population characteristics [Pisprasert et al., 2013]. Accordingly, a one-unit difference in HOMA-IR should not necessarily be assumed to represent an identical physiological difference in insulin sensitivity in women and men. Third, hsCRP is a nonspecific marker of systemic inflammation and does not identify the source or chronicity of the inflammatory process [Pepys and Hirschfield, 2003]. Because hsCRP was measured at a single examination, transient and persistent elevations could not be distinguished. This limitation is particularly relevant given the substantial attenuation of the estimated sex contrast after restriction to $\text{hsCRP} \leq 10 \text{ mg/L}$.

Selection and residual confounding also warrant consideration. The analytic population was deliberately restricted to adults without diagnosed diabetes and with HbA1c below the conventional prediabetes threshold, and the findings therefore should not be assumed to generalize to individuals with prediabetes or diabetes or to populations outside the NHANES target population. Analyses additionally depended on participation in the fasting laboratory subsample and complete availability of primary-model variables. Appropriate fasting-subsample survey weights were used to support population inference, and only 103 of 3,182 otherwise eligible participants were excluded for incomplete primary-model covariates; nevertheless, selection related to measurement availability or complete-case inclusion cannot be excluded. Residual confounding also remains possible despite adjustment for major metabolic covariates and the targeted sensitivity analyses. Race and ethnicity, socioeconomic circumstances, physical activity, diet, reproductive and hormonal factors, additional medication use, and more detailed measures of body composition are among factors not comprehensively represented in the primary model. Their omission should not be interpreted as evidence that they are unimportant, but indiscriminate post hoc adjustment for additional variables would not necessarily reduce bias and could complicate interpretation.

The present analysis should also be viewed as a focused follow-up to an exploratory observation

from an earlier analysis of the same NHANES population rather than as an independent replication. Sex-related heterogeneity was identified during methodological evaluation of the preceding study and motivated the specific HOMA-IR-by-sex interaction evaluated here. The present study subsequently defined that interaction as its primary question and subjected it to structured sensitivity analyses; however, because the hypothesis was generated and evaluated using overlapping data, the primary interaction estimate should not be interpreted as independent confirmation of the original exploratory signal. Replication in an independent population will therefore be important.

The study also has several strengths. The analytic population was deliberately defined to examine metabolic-inflammatory heterogeneity among adults who had not crossed conventional glycemic thresholds for prediabetes or diagnosed diabetes. Sex differences were evaluated using a formal interaction test rather than inferred from comparisons of statistical significance within sex-specific subgroups. Analyses incorporated the complex NHANES sampling design, fasting-subsample weights, and domain-based variance estimation to support population-level inference. The primary interaction was further evaluated across a structured series of sensitivity analyses addressing exposure scale and influence, alternative representations of adiposity, several clinical and behavioral factors, pregnancy, and the upper range of the hsCRP distribution. Where sensitivity analyses altered the available analytic sample, same-sample reference models were used to distinguish changes associated with covariate adjustment from those associated with sample restriction. Finally, sensitivity findings were evaluated on the basis of changes in effect estimates and uncertainty rather than solely according to whether a nominal threshold for statistical significance was crossed.

In conclusion, among U.S. adults without diagnosed diabetes and with HbA1c below the conventional threshold for prediabetes, the adjusted association between HOMA-IR and hsCRP was more positive among women than men. This sex difference persisted across alternative HOMA-IR scales, progressively trimmed exposure distributions, multiple representations of adiposity, and a range of targeted clinical and behavioral sensitivity analyses. However, restriction to $\text{hsCRP} \leq 10 \text{ mg/L}$ reduced the estimated sex contrast by approximately one-half, indicating that the upper portion of the inflammatory distribution contributes importantly to the observed heterogeneity. Taken together, these findings suggest that sex-related differences in metabolic-inflammatory coupling may be detectable even below conventional glycemic thresholds, while also demonstrating that the magnitude of this difference depends substantially on the inflammatory range under study. Independent replication and studies incorporating more direct measures of insulin sensitivity, repeated inflammatory measurements, body composition, and reproductive physiology will be important for determining the biological basis and clinical relevance of this pattern.

Author Contributions

S.M. conceived the study, developed the analytic framework, conducted the statistical analyses, interpreted the results, and drafted the manuscript. J.B. contributed to study development, interpretation of the findings, and critical revision of the manuscript. Both authors reviewed and approved the final manuscript.

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Conflicts of Interest

The authors declare no conflicts of interest.

Ethics Statement

This study used publicly available, de-identified data from the National Health and Nutrition Examination Survey (NHANES). NHANES protocols were approved by the National Center for Health Statistics Research Ethics Review Board, and written informed consent was obtained from participants in accordance with NHANES procedures. Because the present study involved secondary analysis of publicly available, de-identified data, no additional institutional review board approval or informed consent was required.

Data Availability

The NHANES data used in this study are publicly available from the National Center for Health Statistics.

Code Availability

The computational code, aggregate analysis outputs, and supporting validation materials for this study are archived on Zenodo at DOI: [10.5281/zenodo.23200490](https://doi.org/10.5281/zenodo.23200490). The computational repository is also publicly available through GitHub.

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